

FINANCIAL FRAUD DETECTION APPLICATION

Technical Report

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1. INTRODUCTION

Financial fraud has become one of the major challenges in the digital banking and online transaction industry. With the rapid increase in digital payments, cybercriminals use advanced methods to perform unauthorized transactions, identity theft, and financial scams.

The objective of this project is to develop a **Financial Fraud Detection Application** using Machine Learning techniques that can automatically identify suspicious transactions and help financial institutions reduce fraud-related losses.

This system analyzes transaction patterns, calculates fraud probability, generates risk scores, and alerts administrators about suspicious activities.

2. OBJECTIVES

The main objectives of this project are:

- To detect fraudulent financial transactions using Machine Learning.
- To automate fraud monitoring and reduce manual efforts.
- To generate risk scores for suspicious transactions.
- To provide real-time transaction analysis and visualization.
- To create an interactive dashboard for fraud analytics.

3. PROBLEM STATEMENT

Traditional fraud detection systems rely heavily on manual verification and rule-based systems. These approaches are slow, less accurate, and difficult to scale for large transaction volumes.

The Financial Fraud Detection App solves this problem by:

- Using Machine Learning algorithms for intelligent fraud prediction.
- Processing large transaction datasets efficiently.
- Providing faster and more accurate fraud detection.
- Supporting real-time analytics and monitoring.

4. TECHNOLOGY STACK

Sr. No.	Technology / Tool	Category	Purpose
1	Python	Programming Language	Core development language used for backend and ML implementation
2	Streamlit	Frontend Framework	Used to create the interactive dashboard and user interface
3	Pandas	Data Processing	Used for data cleaning, manipulation, and preprocessing
4	NumPy	Numerical Computing	Used for mathematical and numerical operations
5	Scikit-learn	Machine Learning	Used for model training, preprocessing, and evaluation
6	XGBoost	Machine Learning Algorithm	Used for advanced fraud prediction with high accuracy
7	Plotly	Data Visualization	Used for graphs, charts, and dashboard analytics
8	SQLAlchemy	ORM / Database Connectivity	Used for database management and ORM operations
9	SQLite	Database	Used for storing transaction and user data locally
10	MySQL	Database	Used for scalable database storage in production
11	VS Code	Development Environment	Used for coding, debugging, and development

5. SYSTEM ARCHITECTURE

The application consists of multiple modules:

5.1 USER INTERFACE

- Built using Streamlit.
- Provides dashboard and prediction interface.
- Displays fraud analytics and reports.

5.2 ETL PIPELINE

- Extracts and preprocesses transaction data.
- Handles missing values.
- Encodes categorical data.
- Normalizes features using MinMaxScaler.

5.3 MACHINE LEARNING MODULE

The system trains multiple models:

- Logistic Regression
- Random Forest
- XGBoost

The best model is selected based on F1 Score and prediction accuracy.

5.4 FRAUD PREDICTION MODULE

- Predicts fraud probability.
- Generates risk scores from 0–100.
- Categorizes transactions into:
 - Low Risk
 - Medium Risk
 - High Risk
 - Critical Risk

5.5 DATABASE MODULE

- Stores transaction records.
- Maintains user login information.

- Supports SQLite and MySQL.

6. WORKING OF THE SYSTEM

The workflow of the application is as follows:

1. User enters transaction details.
2. Data is pre-processed using the ETL pipeline.
3. Features are transformed and normalized.
4. The trained Machine Learning model predicts fraud probability.
5. Risk score is generated.
6. Result is displayed on dashboard.
7. Transaction data is stored in database.

7. FEATURES OF THE APPLICATION

- Fraud prediction using Machine Learning.
- Interactive dashboard with visual analytics.
- Real-time transaction monitoring.
- Risk score calculation.
- CSV batch transaction scoring.
- Authentication system with secure password hashing.
- Email alert generation.
- Database integration.
- Live analytics and reporting.

8. MACHINE LEARNING MODELS USED

Logistic Regression

A supervised learning algorithm used for binary classification problems.

Random Forest

An ensemble learning algorithm that improves prediction accuracy using multiple decision trees.

XGBoost

An advanced boosting algorithm known for high performance and efficiency in fraud detection tasks.

9. RISK SCORE FORMULA

The fraud probability generated by the model is converted into a risk score.

Risk Level	Score Range
Low	0 – 25
Medium	26 – 50
High	51 – 75
Critical	76 – 100

10. CONCLUSION

The Financial Fraud Detection Application successfully demonstrates how Machine Learning can be used to identify suspicious financial transactions efficiently.

The system provides automated fraud prediction, risk analysis, alert generation, and real-time monitoring through an interactive dashboard. This project helps improve financial security and reduces fraud-related risks for organizations.

Through this project, practical knowledge of Machine Learning, data preprocessing, dashboard development, database integration, and real-time analytics was gained.